

RESEARCH NOVELTY STATEMENT

Course Submission Document

Course: Research on Data Analysis
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Instructor: Assoc. Prof. Dr. Nguyen Dinh Thuan
Paper title: Are Counterfactual HR Intervention Scenarios Feasible? A Pilot Audit of Employees with High Predicted Attrition Risk
Student name: Nguyen Tien Khoa
Student ID: 250104039
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1. Research gap

Existing employee-attrition studies commonly address one or more of three tasks: predicting which employees are at risk, explaining predictions through feature attribution, and generating counterfactual profiles that move a model prediction from high risk to low risk. These tasks do not by themselves establish whether a score-reducing profile represents an action that HR can reasonably evaluate. A counterfactual may change age, work history, tenure, job structure, or a satisfaction score. Such changes can be valid within a model while remaining immutable, retrospective, indirect, logically inconsistent, or outside the scope of a short-term HR intervention.

Prior employee-attrition studies already use feature constraints, mutability rules, and counterfactual explanations to derive retention-oriented insights. The gap addressed here is therefore narrower: existing components do not by themselves provide a transparent audit that quantifies, for every flagged employee, how apparent recourse coverage changes when the search is restricted to a small predeclared HR intervention space; why scenarios are rejected; and whether a missing result reflects substantive no-recourse or a technical failure.

2. Novelty of the study

The novelty of this study lies in the analytical design and the form of its decision-support output:

1. **Operational separation of two concepts.** The study distinguishes a *model-valid scenario*, which only crosses the model threshold, from an *HR-rule-feasible scenario*, which must also satisfy predefined rules for scope, direction, range, salary limits, immutability, and logical consistency.
2. **Three-level coverage-loss audit.** Broad and intermediate DiCE searches are retained as diagnostic baselines, while the main analysis restricts changes to MonthlyIncome, StockOptionLevel, and OverTime. The bounded Level 3 grid is exhaustively evaluated. This makes the loss of apparent actionability after domain restrictions measurable rather than anecdotal; the novelty claim is not that constrained counterfactuals themselves are new.
3. **Employee-level accountability.** Instead of presenting only a few favorable counterfactual examples, the framework produces an assessment for every high-risk employee. It records whether recourse exists, whether a Level 3 scenario passes the encoded rules, which scenario

is selected under a transparent ranking policy, and why alternatives are rejected. Technical failures are kept separate from substantive no-recourse findings.

4. **Assumption-aware feasibility analysis.** The study examines whether the feasibility conclusion changes across data splits and under alternative risk thresholds, salary-cap percentiles, and overtime policies. The output therefore describes not only estimated coverage but also which operational assumptions drive it.

3. Difference from a conventional attrition analysis

Conventional approach	Contribution of this study
Predict who is likely to leave	Determine whether the model exposes any bounded intervention pathway for each flagged employee
Explain important features	Separate predictive attribution from variables that can be treated as direct actions
Generate score-changing counterfactuals	Audit scenarios against explicit HR rules and report rejection reasons
Present selected examples	Produce an assessment and final group for every successfully analyzed high-risk employee
Treat feasibility as a fixed result	Test sensitivity to the split and to operational assumptions

4. Empirical evidence supporting the contribution

In the primary test split, the validation-selected model identified 117 high-risk employees. Broad and intermediate searches found model-valid diagnostic recourse for all 117 employees. After the intervention space was restricted to the declared HR levers and rules, only 73 employees had at least one Level 3 scenario that crossed the model threshold and passed the encoded audit. The remaining 44 employees had diagnostic model recourse but no feasible scenario in the main intervention space.

Result	Employees	Rate
Diagnostic model-valid recourse	117/117	100.0%
Level 3 rule-feasible recourse	73/117	62.4%
Model-valid but not HR-rule-feasible	44/117	37.6%
Level 3 recourse when overtime is locked	42/117	35.9%

The central empirical finding is the 37.6-percentage-point gap between diagnostic computational recourse and feasibility under the encoded HR intervention rules. The overtime sensitivity result further shows that the estimated intervention coverage depends materially on whether workload and scheduling can be changed.

5. Scope of the novelty claim

This study does **not** claim to introduce a new machine-learning classifier, a new SHAP method, a new counterfactual-generation algorithm, or the general idea of constraining mutable features.

Related employee-attrition work by Artelt and Gregoriades (2023) already used feature blacklists and constraints, while Lv et al. (2026) combined mutability-constrained counterfactuals with strategy generation and sensitivity analysis. The contribution claimed here is narrower: a reproducible three-level audit that measures coverage loss, reports employee-level feasibility and rejection reasons, preserves failure semantics, and tests the assumptions governing a bounded HR grid.

The IBM dataset is synthetic and cross-sectional. Consequently, a rule-feasible scenario means feasible only under the encoded study rules; it is not an approved HR action and does not establish that the intervention would causally prevent attrition. The framework should be interpreted as a model-audit and structured human-review tool rather than an automated retention decision system.

6. Selected references

The first two studies below are the closest employee-attrition comparators and are used to delimit the novelty claim. The remaining works provide the methodological foundations for diverse counterfactuals, actionable recourse, feasibility, and domain-guided constraints.

1. Artelt, A., and Gregoriades, A. (2023). “How to Make Them Stay?: Diverse Counterfactual Explanations of Employee Attrition.” *ICEIS 2023*. doi:10.5220/0011961300003467.
2. Lv, C., Chen, Y., Li, H., and Dong, Y. (2026). “A Decision-Oriented Approach for Employee Turnover Prediction with Counterfactual Explanations.” *Expert Systems with Applications*. doi:10.1016/j.eswa.2026.133494.
3. Mothilal, R. K., Sharma, A., and Tan, C. (2020). “Explaining Machine Learning Classifiers through Diverse Counterfactual Explanations.” *FAccT 2020*. doi:10.1145/3351095.3372850.
4. Ustun, B., Spangher, A., and Liu, Y. (2019). “Actionable Recourse in Linear Classification.” *FAT* 2019*. doi:10.1145/3287560.3287566.
5. Poyiadzi, R., Sokol, K., Santos-Rodriguez, R., De Bie, T., and Flach, P. (2020). “FACE: Feasible and Actionable Counterfactual Explanations.” *AIES 2020*. doi:10.1145/3375627.3375850.
6. Karimi, A. H., Barthe, G., Schölkopf, B., and Valera, I. (2022). “A Survey of Algorithmic Recourse: Contrastive Explanations and Consequential Recommendations.” *ACM Computing Surveys*. doi:10.1145/3527848.
7. Thieu, V. (2026). “Knowledge-Guided Counterfactual Explanations for Diabetes Risk Decision Support: A Directional Intervention Taxonomy.” *International Journal of Medical Informatics*, 219, 106555. doi:10.1016/j.ijmedinf.2026.106555.